

Weijie Shen

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SUMMARY

C++-focused software engineer with 6+ years across backend systems, analytics tooling, and performance-sensitive applications. Main stack is C++11/14/17/20, with Python and some fullstack work when the product needs it. Most projects have involved code that has to stay fast, testable, and maintainable across large codebases.

EXPERIENCE

- **OpenText** Waterloo, ON
Senior Software Engineer (Full Stack) Apr 2025 – Present
 - **AI Log Analysis & RCA Automation:** Built a Copilot subagent workflow that pulls Jira tickets and related documents, filters them with team domain knowledge, runs AI-assisted log analysis, and sends RCA drafts to customer support. It is being shaped into a VM/cloud-hosted web service so support can handle common cases without waiting on developers.
 - **System Development:** Build and maintain C++ backend services and frontend administrative tools used around business-critical workflows.
 - **Design Collaboration:** Work with architects and product managers to sort out design dependencies before implementation. Complete high level and low level design document for monitoring system using grafana and prometheus.
- **Ansys (Synopsys)** Vancouver, BC
R&D Software Engineer II Oct 2021 – Apr 2025
 - **Analytics Algorithm Development in C++:** Built analytics algorithms in modern C++ for GPU architecture evaluation, with strict latency and throughput targets on large datasets.
 - **Architecture & Code Quality:** Refactored 1.5k source files into a decoupled MVC architecture using C++17 OOP patterns; Take ownership of untouched module and reduced regressions and bugs by 60%.
 - **Automated Testing & Cross-Platform Debugging:** Wrote automated regression and performance tests in Python and tracked down multi-platform defects across large codebases.
- **LifeWorks (Telus Health)** Montreal, QC
Software Analyst (Backend) Jun 2020 – Jun 2021
 - **Performance Optimization & Code Quality:** Reduced system processing time by 15% through targeted algorithmic optimizations; led code reviews during production migration to C#/.NET Core.
- **Autolog** Montreal, QC
Computer Vision Engineer Jan 2019 – Feb 2020
 - **Object Detection in C++:** Wrote a multi-threaded object detection algorithm in C++ for real-time throughput.

TECHNICAL SKILLS

- **Languages & Standards:** C++ (C++11/14/17/20), Python, TypeScript, JavaScript, Angular, Java, C#, Rust; OOP, design patterns, data structures & algorithms
- **Tools & Practices:** Git, Bitbucket, Jira, Bitbucket, Confluence, MS Visual Studio, VS Code, SonarQube, GitHub CoPilot, CICD, Agile/Scrum, test automation, performance profiling

TECHNICAL PROJECTS

- **Multithreaded Server (Windows & Linux):** C++ server built to run the same way on Windows and Linux, with unit and integration tests covering both platforms.
- **AWS Inventory Management:** Next.js/Node.js inventory application with Redux caching and AWS EC2/S3/RDS for persistence. Added automation around data sync and workflow-heavy tasks.

EDUCATION

- **École Polytechnique de Montréal** Montreal, QC
Master of Engineering 2018